



Short Communication

Long-term cancer and overall mortality associated with drinking water nitrate in the United States

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ABSTRACT

Objectives: Nitrate is a probable carcinogen regulated in drinking water by the US Environmental Protection Agency (EPA) to a maximum contaminant level (MCL) of 10 mg/L nitrate-nitrogen (NO₃-N; equivalent to 44.3 mg/L NO₃). We aimed to determine the association of US drinking water nitrate levels with overall as well as cardiovascular, cancer, and other cause mortality.

Study design: This study used a population-based retrospective cohort design.

Methods: We analyzed data from 2029 participants of the 2005–2006 National Health and Nutrition Examination Survey followed for mortality until 2019 for a median of 13.9 years. We used Cox proportional hazards regression to estimate the hazard ratio (HR) and 95% confidence interval (CI) for mortality associated with drinking water nitrate, adjusting for covariates that included socio-economic factors and pack-years of cigarette smoking.

Results: Drinking water nitrate was detected in 50.8 % of the samples, had a median concentration of 0.77 mg/L NO₃, and was above US EPA MCL in 0.4 % of participants. In adjusted analysis, drinking water nitrate detection was associated with 73 % higher cancer mortality (HR: 1.73, 95% CI: 1.19–2.51), whereas a 10-fold increase in drinking water nitrate levels was associated with 69 % higher cancer mortality (HR: 1.69, 95% CI: 1.24–2.31) and 21 % higher overall mortality (HR: 1.21, 95% CI: 1.00–1.46). Drinking water nitrate below EPA MCL was still associated with higher cancer mortality (HR: 1.61, 95% CI: 1.07–2.43 per 10-fold increase and HR: 1.61, 95% CI: 1.08–2.42 for detection).

Conclusions: Levels of drinking water nitrate may be an overlooked contributor to cancer mortality in the United States.

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Introduction

Nitrate is a common regulated chemical in drinking water.¹ The use of fertilizers in agriculture and the application of manure from concentrated animal feeding operations (CAFO) are the main sources of nitrate, which can accumulate in soil and contaminate both surface water and groundwater.¹ In the United States, nitrate is present at low concentrations (<4 mg/L nitrate [NO₃]) in most surface water and groundwater supplies but can exceed 20 mg/L NO₃ in 3 %–6 % of surface and ground waters.² Over the recent years, drinking water nitrate has increased in many US states due to

higher nitrogen input from fertilizer application and CAFOs and to ammonium deposition.³ In Europe, up to 10 % of the populations from 15 countries are exposed to nitrate levels in drinking water above the European Union regulatory limit of 50 mg/L NO₃.² Nitrate can persist for decades in groundwater under aerobic conditions.⁴

The US Environmental Protection Agency (EPA) has set the current maximum contaminant level (MCL) for drinking water nitrate to 10 mg/L nitrate-nitrogen (NO₃-N; equivalent to 44.3 mg/L NO₃) in the prevention of methemoglobinemia, a condition characterized by a form of hemoglobin with ferric iron unable to deliver oxygen in infants.⁵ However, this MCL did not consider other health effects such as cancers of the digestive and genitourinary systems as well as adverse reproductive outcomes reportedly associated with drinking water nitrate.^{6,7} Therefore, our objective was to test the hypothesis that drinking water nitrate levels in the United States are associated with higher risk of mortality and to identify the specific causes of death related to the exposure.

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Methods

We used data from the National Health and Nutrition Examination Surveys (NHANES) conducted in 2005–2006 by the National Center for Health Statistics of the Centers for Disease Control and Prevention. The NHANES included 3261 participants aged ≥12 years who had data on drinking water nitrate. Among them, only adults aged ≥20 years had data on mortality and all covariates (*n* = 2029) and were included in our analysis.⁸ Tap water nitrate was quantified at the National Center for Environmental Health using ion chromatography coupled with electrospray tandem mass spectrometry and the limit of detection (LOD) was 0.70 mg/L NO₃. NHANES participants were matched to National Death Index records and using death certificates for confirmation. The specific causes of mortality in our analysis included mortality from cardiovascular disease (diseases of the heart and stroke) and cancer, defined using a standardized list of 113 causes according to the Tenth Revision of the International Classification of Diseases. The low numbers of deaths from other causes precluded any further stratification. The NHANES also collected data on age, gender, race/ethnicity, family poverty income ratio (PIR), cigarette smoking, diabetes, and hypertension. Body mass index (BMI) was calculated as weight in kilograms divided by height in meters squared. NHANES participants were also asked about their source of tap water (private/public company, private/public well, or other) and the use of any water treatment device in their home. We performed descriptive analyses to summarize the characteristics of study participants and used Cox proportional hazards regression to estimate the hazard ratio (HR) and corresponding 95% confidence interval (CI) for the risk of mortality associated with drinking water nitrate detection as well as log₁₀ transformed concentrations. The models were adjusted for age, PIR, BMI, and pack-years of cigarette smoking used as continuous variables as well as sex, race/ethnicity, diabetes, hypertension, source of tap water, and the use of a water treatment device used as categorical

variables. We assessed the proportional hazards assumption using Schoenfeld residuals, and we found no violations. The complex survey design and the sampling weights were accounted for in all analyses to obtain unbiased national estimates. The analyses were performed in R version 4.3.1, and *P*-values <0.05 were considered statistically significant.

Results

Our sample consisted of 2029 participants with a median follow-up of 13.9 years, during which 364 deaths occurred, including 127 from cardiovascular disease, 75 from cancer, and 102 from other causes. At baseline, water nitrate was detected in 50.8 % of the samples; it had a median of 0.77 mg/L NO₃ (interquartile range: <LOD, 3.67) and ranged from <LOD to 80.50 mg/L NO₃. Drinking water nitrate was above US EPA MCL (>44.3 mg/L NO₃) in 0.4 % of samples. Participants aged 20–39 years, non-smokers, and those who had a private/public company source of tap water had higher prevalence of nitrate detection in drinking water (Supplemental Table). In Cox proportional regression adjusted for covariates, 10-fold increase in drinking water nitrate was associated with 21 % higher all-cause mortality (HR: 1.21, 95% CI: 1.00–1.46). Cancer mortality was higher by 73 % with detection nitrate in drinking water (HR: 1.73, 95% CI: 1.19–2.51) and by 69 % with 10-fold increase in drinking water nitrate (HR: 1.69, 95% CI: 1.24–2.31). No association was found between the exposure and cardiovascular mortality or mortality due to other causes (Fig. 1). In an analysis restricted to participants with drinking water nitrate levels below US EPA MCL, drinking water nitrate levels (HR: 1.61, 95% CI: 1.07–2.43 per 10-fold increase) and detection (HR: 1.61, 95% CI: 1.08–2.42) were still associated with higher cancer mortality. The association of drinking water nitrate levels with overall mortality was attenuated and became marginally significant (*P*-value: 0.09; HR: 1.20, 95% CI: 0.97–1.49 per 10-fold increase).

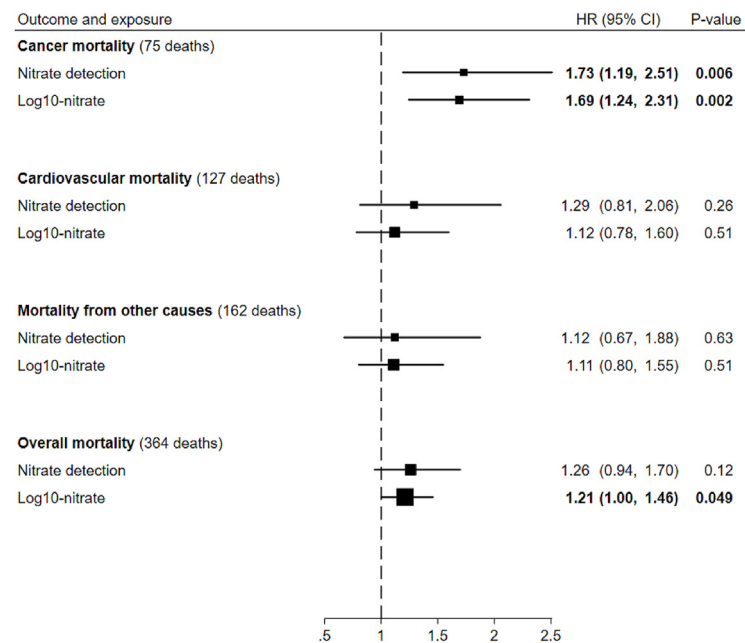


Fig. 1. Forest plots for the association of drinking water nitrate with mortality from cancer, cardiovascular diseases, other causes, and overall. Hazard ratios (HRs) and 95% confidence intervals (CIs) calculated using Cox proportional hazards regression. Models adjusted for age, gender, race/ethnicity, poverty income ratio (PIR), body mass index (BMI), pack-years of cigarette smoking, diabetes, hypertension, source of tap water, and use of a water treatment device. Bold values indicate significant associations.

Discussion

Our results suggest that drinking water nitrate levels in the United States are associated with a higher risk of cancer mortality, even at levels below US EPA MCL, and overall mortality. There have been no previous studies on water nitrate's association with overall mortality; however, our findings on drinking water nitrate and cancer mortality are consistent with previously reported associations of drinking water nitrate with higher risk of digestive and urogenital cancer.^{6,7} Moreover, there are some reports of prostate, breast, and thyroid cancers, non-Hodgkin lymphoma as well as brain tumors associated with drinking water nitrate.^{6,7,9} The mechanisms by which nitrate could lead to cancer have previously been described. Nitrate ingested through drinking water can be reduced by oral bacteria to nitrite, which enters the enterosalivary circulation to be transformed into N-nitroso compounds, which are known human carcinogens.⁸ Mechanisms for carcinogenicity include DNA adducts formation and DNA methylation that can activate proto-oncogenes.¹⁰

Our analysis had limitations. We did not have information on cancer incidence and mortality from specific cancer types. We cannot rule out that water nitrate might be a marker for other carcinogenic water contaminants. We did not have enough power to examine mortality from other causes, and we lacked information on the duration of exposure. We also had no information on dietary intake of nitrate or nitrite. However, our analysis included a sample representative of the US population, which contributes to the generalizability of the findings and reduces selection bias. In conclusion, this analysis provides further evidence that drinking water nitrate may be a risk factor for mortality from cancer.

Warranted the associations observed in the present analysis are causal, measures to reduce nitrate inputs to surface and ground water resources as well as nitrate removal systems for finished water treatment may reduce mortality, especially from cancer.

Author statements

Ethical approval

None required.

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Competing interests

The authors have no disclosure related to the submitted manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.puhe.2024.01.001>.

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